

APTITUDE INTERACTIVE COMPLETE HANDBOOK

Number System • Percentage • Profit & Loss • SI & CI • Ratio & Proportion • Average
• Time & Work • TSD • Probability • P&C; • DI

HOW TO USE THIS BOOK

Read the concept cards → attempt the Quick Checks → learn the Speed Tricks → watch the Exam Traps → solve 20 questions per chapter → check the answer key → log mistakes in the Error Log.

Designed for self-study: spacious layout, highlighted essentials, practice space and progressive difficulty.

CONTENTS

Chapter 01 — Number System

Basic → Advanced: build the foundation before speed.

Chapter 02 — Percentage

Conversions, change, comparison, reverse percentage and exam shortcuts.

Chapter 03 — Profit & Loss

CP/SP/MP, discounts, successive changes and dishonest-dealer traps.

Chapter 04 — Simple & Compound Interest

SI, CI, depreciation, compounding frequency and comparison.

Chapter 05 — Ratio & Proportion

Ratios, mixtures, variation, proportional division and shortcuts.

Chapter 06 — Average

Core average, weighted/combined average, replacement and speed.

Chapter 07 — Time & Work

Rates, efficiency, LCM method, pipes and changing teams.

Chapter 08 — Time, Speed & Distance

Relative speed, trains, boats, average speed and catch-up.

Chapter 09 — Probability

Sample space, complements, cards, dice, conditional and dependent events.

Chapter 10 — Permutation & Combination

Counting, restrictions, circular arrangements, repetition and distributions.

Chapter 11 — Data Interpretation

Tables, charts, percentages, ratios, averages, profit and mixed DI.

MASTER CHECKLIST

- I understand the formulas & concepts
- I can solve the Quick Checks without help
- I finished all 20 questions
- I reviewed every wrong answer
- I added recurring mistakes to the Error Log

Number System

Basic → Advanced: build the foundation before speed.

■ LEARNING TARGET

By the end of this chapter you should be comfortable with divisibility, factors, HCF/LCM, remainders, modular arithmetic, unit digits, factorials, perfect powers, counting multiples and advanced congruences.

■ CORE ROADMAP

- Number families: Natural (1,2,3...), Whole (0,1,2...), Integers (...,-1,0,1...), Rational p/q , Irrational and Real numbers.
- Prime = exactly two positive factors. Composite = more than two. 1 is neither prime nor composite.
- Co-prime numbers have HCF = 1; they need not both be prime.

■ FACTORS & DIVISORS

- If $n = p^a q^b r^c$, number of positive divisors = $(a+1)(b+1)(c+1)$.
- Sum of divisors = $[(p^{a+1}-1)/(p-1)] \times \dots$
- Number of odd divisors: ignore exponent of 2. Number of even divisors = total – odd divisors.
- A perfect square has even exponents in prime factorization; a perfect cube has exponents divisible by 3.

■ DIVISIBILITY

- 2: last digit even. 3/9: digit sum divisible by 3/9. 4: last two digits divisible by 4. 8: last three digits divisible by 8.
- 5: last digit 0/5. 10: last digit 0. 6: divisible by 2 and 3. 12: divisible by 3 and 4.
- 11: alternating digit-sum difference is 0 or a multiple of 11. 25: last two digits 00,25,50,75. 125: last three digits divisible by 125.
- 15 = divisible by 3 and 5; 18 = divisible by 2 and 9; 20 = divisible by 4 and 5.

■ HCF & LCM

- Using prime factors: HCF takes minimum powers; LCM takes maximum powers.
- For two positive integers a, b : $a \times b = \text{HCF}(a, b) \times \text{LCM}(a, b)$.
- Euclidean algorithm: $\text{HCF}(a, b) = \text{HCF}(b, a \bmod b)$.
- For fractions: $\text{HCF} = \text{HCF}(\text{numerators}) / \text{LCM}(\text{denominators})$; $\text{LCM} = \text{LCM}(\text{numerators}) / \text{HCF}(\text{denominators})$, after reduction.

■ REMAINDERS & MODULAR ARITHMETIC

- Division algorithm: $N = dq + r$, where $0 \leq r < d$.
- $(a+b) \bmod m$ and $(ab) \bmod m$ can be evaluated using remainders first.
- For $N \equiv r \pmod{m}$, every N has the form $mk+r$.
- To solve simultaneous congruences, look for a common residue; for coprime moduli, CRT gives a unique residue modulo their product.

■ UNIT DIGIT & LAST DIGITS

- Find the cycle of the base. Example: powers of 7 have unit-digit cycle 7,9,3,1 of length 4.
- For last two/three digits, work modulo 100/1000 and use repeated squaring or cycles.
- If a number ends in 01, its powers often stay 01 modulo 100; always verify the modulus.

■ FACTORIAL & TRAILING ZEROES

- Highest power of prime p in $n!$: $\text{floor}(n/p) + \text{floor}(n/p^2) + \text{floor}(n/p^3) + \dots$
- Trailing zeroes in $n!$ = highest power of 10 = $\min(v_2, v_5)$; normally v_5 is smaller.
- Trailing zeroes in $n!$ = $\text{floor}(n/5) + \text{floor}(n/25) + \text{floor}(n/125) + \dots$

■ SEQUENCES & SUMS

- $1+2+\dots+n = n(n+1)/2$. First n odd numbers sum to n^2 . First n even numbers sum to $n(n+1)$.
- Arithmetic progression: $a_n = a + (n-1)d$; $S_n = n/2[2a + (n-1)d]$.
- Number of multiples of k from 1 to N = $\text{floor}(N/k)$. For 'A or B', use inclusion-exclusion.

■ ADVANCED NUMBER THEORY

- Euler phi: $\phi(n) = n \prod (1 - 1/p)$ over distinct prime divisors p of n .
- Euler theorem: if $\gcd(a, n) = 1$, $a^{\phi(n)} \equiv 1 \pmod{n}$.
- Wilson theorem: for prime p , $(p-1)! \equiv -1 \pmod{p}$.
- Useful patterns: odd $n \Rightarrow n^2 \equiv 1 \pmod{8}$. Prime $p > 3 \Rightarrow p^2 \equiv 1 \pmod{24}$.

FORMULA CARD

Divisors $n = p^a q^b r^c \Rightarrow d(n) = (a+1)(b+1)(c+1)$

HCF/LCM For two numbers: $\text{product} = \text{HCF} \times \text{LCM}$

Trailing zeroes $v_5(n!) = \text{floor}(n/5) + \text{floor}(n/25) + \dots$

Euler phi $\phi(n) = n \prod (1 - 1/p)$

■ QUICK CHECK

What is the number of divisors of 360?

Answer: 24

■ QUICK CHECK

What is $7^{100} \bmod 100$?

Answer: 01

■ SPEED TRICK

For big powers, never expand. Reduce the base modulo the divisor, identify the cycle, then reduce the exponent.

■ EXAM TRAP

1 is neither prime nor composite. Also, 'number of factors' means positive divisors unless stated otherwise.

Number System — 20 QUESTION PRACTICE SET

Difficulty is progressive: Easy 1–7 • Medium 8–14 • Hard 15–20

■ EASY

Solve without looking at the answer key. Mark one ✓ when done.

1. Find HCF of 36 and 48.

2. Find LCM of 12 and 18.

3. Is 1 prime, composite, or neither?

4. Find $12345 \bmod 9$.

5. How many positive divisors does 72 have?

6. Find the unit digit of 7^{25} .

7. Find $1+2+\dots+50$.

■ MEDIUM

Solve without looking at the answer key. Mark one ✓ when done.

8. Find the least positive N such that N leaves remainder 3 when divided by 5, remainder 6 when divided by 8, and remainder 6 when divided by 9.

9. Find the highest power of 5 dividing $100!$.

10. Find the number of trailing zeroes in $125!$.

11. Find $3^{100} \bmod 7$.

12. How many integers from 1 to 500 are divisible by 3 or 5?

13. Find the smallest perfect square divisible by 12, 18 and 30.

14. Find the sum of all positive divisors of 360.

■ **HARD**

Solve without looking at the answer key. Mark one ✓ when done.

15. Find $(2^{1000} + 3^{1000}) \bmod 7$.

16. How many perfect-square divisors does $2^6 \times 3^4 \times 5^2$ have?

17. Find the last two digits of 7^{100} .

18. Find the least positive N satisfying $N \equiv 2 \pmod{3}$, $N \equiv 3 \pmod{4}$, $N \equiv 4 \pmod{5}$.

19. Find the highest power of 3 dividing $1000!$.

20. How many integers from 1 to 1000 are coprime to 30?

Number System — ANSWER KEY & MINI REVIEW

1. 12	2. 36	3. Neither	4. 6	5. 12
6. 7	7. 1275	8. 78	9. 24	10. 31
11. 4	12. 233	13. 900	14. 1170	15. 6
16. 24	17. 01	18. 59	19. 498	20. 266

■■ 20/20 TRACKER

Easy: ■1 ■2 ■3 ■4 ■5 ■6 ■7

Medium: ■8 ■9 ■10 ■11 ■12 ■13 ■14

Hard: ■15 ■16 ■17 ■18 ■19 ■20

■ ERROR LOG — WRITE YOUR OWN PATTERN

Question no.: _____ Topic: _____

My mistake: _____

Correct idea/formula: _____

How I will avoid it next time: _____

Percentage

Conversions, change, comparison, reverse percentage and exam shortcuts.

■ CORE CONCEPT

- $x\% = x/100$. Convert percentages to fractions early.
- Percentage change = $(\text{change}/\text{original}) \times 100$.
- Reverse percentage: if final = 120% of original, original = final/1.2.

■ SUCCESSIVE CHANGE

- Two changes $a\%$ and $b\%$ give net % = $a+b+ab/100$ when both are increases.
- For increase $a\%$ followed by decrease $b\%$, net = $a-b-ab/100$.
- A decrease of $x\%$ followed by an increase of $x\%$ is always a net loss $x^2/100\%$.

■ COMPARISON & REVERSE

- If A is $x\%$ more than B, $A:B = (100+x):100$.
- If A is $x\%$ less than B, $A:B = (100-x):100$.
- If A is $x\%$ more than B, then B is $100x/(100+x)\%$ less than A.

■ APPLICATIONS

- Marks, votes, population, salary, expenditure, income and error questions are usually direct percentage models.
- Percentage error = $|\text{measured} - \text{true}|/\text{true} \times 100$.
- Always identify the base: '% of what?'

FORMULA CARD

Change $\% \text{ change} = \text{change}/\text{original} \times 100$

Successive $a+b+ab/100$ for two increases

Reverse $\text{Original} = \text{final}/\text{multiplier}$

■ QUICK CHECK

A 20% increase followed by 20% decrease gives what net change?

Answer: 4% decrease

■ QUICK CHECK

If A is 25% more than B, B is what % less than A?

Answer: 20%

■ SPEED TRICK

Use multipliers: $+20\% = \times 1.20$, $-20\% = \times 0.80$. This makes successive changes almost automatic.

■ EXAM TRAP

The base changes in comparison statements. 'A is 20% more than B' does not mean B is 20% less than A.

Percentage — 20 QUESTION PRACTICE SET

Difficulty is progressive: Easy 1–7 • Medium 8–14 • Hard 15–20

■ EASY

Solve without looking at the answer key. Mark one ✓ when done.

1. Find 25% of 480.

2. 30 is what percent of 100?

3. Increase 400 by 40%.

4. 75 is what percent of 500?

5. A number is 20% less than 400. Find it.

6. 80 increased to 137.6. Find the percentage increase.

7. A price of 200 rises by 15%. Find the new price.

■ MEDIUM

Solve without looking at the answer key. Mark one ✓ when done.

8. A number increases by 20% and then decreases by 20%. Find the net percentage change.

9. If 40% of x equals 80, find x .

10. A is 20% less than B. By what percent is B more than A?

11. 40% of $x = 25\%$ of y . Find $x:y$.

12. A value rises from 200 to 253. Find percentage increase.

13. A number is increased by 25% to become 500. Find the original.

14. A price falls from 250 to 180. Find percentage decrease.

■ **HARD**

Solve without looking at the answer key. Mark one ✓ when done.

15. A value falls by 10% twice. Find total percentage decrease.

16. Income rises 25% and expenditure rises 10%. If initially expenditure was 80% of income, find the percentage increase in savings.

17. A student gets 45% and misses pass by 18 marks. At 54%, he is 18 marks above pass. Find maximum marks.

18. In an election, winner gets 56% and wins by 1440 votes. Find total votes.

19. If $(1+x)^2=1.44$ and $x>0$, find x .

20. 60% of $x = 48\%$ of y . Find $x:y$.

Percentage — ANSWER KEY & MINI REVIEW

1. 120	2. 30%	3. 560	4. 15%	5. 320
6. 72%	7. 230	8. 4% decrease	9. 200	10. 25%
11. 5:8	12. 26.5%	13. 400	14. 28%	15. 19%
16. 85%	17. 400	18. 12000	19. 20%	20. 4:5

■■ 20/20 TRACKER

Easy: ■1 ■2 ■3 ■4 ■5 ■6 ■7

Medium: ■8 ■9 ■10 ■11 ■12 ■13 ■14

Hard: ■15 ■16 ■17 ■18 ■19 ■20

■ ERROR LOG — WRITE YOUR OWN PATTERN

Question no.: _____ Topic: _____

My mistake: _____

Correct idea/formula: _____

How I will avoid it next time: _____

Profit & Loss

CP/SP/MP, discounts, successive changes and dishonest-dealer traps.

■ CORE CONCEPT

- Profit = $SP - CP$; Loss = $CP - SP$.
- Profit% = $\text{Profit}/CP \times 100$; Loss% = $\text{Loss}/CP \times 100$.
- $SP = CP(1 \pm \text{rate}/100)$.

■ DISCOUNT & MARKED PRICE

- Discount = $MP - SP$; Discount% = $\text{Discount}/MP \times 100$.
- $SP = MP(1 - d/100)$.
- Successive discounts a, b: equivalent discount = $a + b - ab/100$.

■ REVERSE & TRAPS

- If SP is 20% below CP, required increase to return to CP is $20/80 = 25\%$.
- If a dealer gives 900g for the price of 1kg, effective gain on quantity is $100/900 \times 100$.
- Equal SP with equal gain/loss percentages does not mean equal CP.

■ EXAM MODELS

- Markup then discount: $SP = CP(1 + \text{markup})(1 - \text{discount})$.
- For 'same SP, one gain and one loss', calculate each CP separately before combining.
- Keep CP as the base for profit/loss; MP is the base for discount.

FORMULA CARD

Profit $SP - CP$

Profit% $\text{Profit}/CP \times 100$

Discount% $(MP - SP)/MP \times 100$

■ QUICK CHECK

CP=800, profit=15%. SP?

Answer: 920

■ QUICK CHECK

20% and 10% successive discounts = ?

Answer: 28%

■ SPEED TRICK

Write CP=100 whenever only percentages are given. Then calculate SP/MP with multipliers.

■ EXAM TRAP

Profit% is on CP; discount% is on MP. Mixing the bases is the classic trap.

Profit & Loss — 20 QUESTION PRACTICE SET

Difficulty is progressive: Easy 1–7 • Medium 8–14 • Hard 15–20

■ EASY

Solve without looking at the answer key. Mark one ✓ when done.

1. CP=500, SP=600. Find profit%.

2. CP=800, loss=10%. Find SP.

3. SP=1020 at 20% profit. Find CP.

4. CP=900, loss=150. Find loss%.

5. MP=1000, discount=15%. Find SP.

6. CP=240, profit=20%. Find SP.

7. SP=990 at 10% loss. Find CP.

■ MEDIUM

Solve without looking at the answer key. Mark one ✓ when done.

8. An item is sold at 12.5% profit for 900. Find CP.

9. CP=1000, MP=1250, discount=20%. Find profit%.

10. A shopkeeper marks up 40% and gives 15% discount. Find profit%.

11. An article sold at 20% loss must be increased by what percent to return to CP?

12. The CP of 12 articles equals the SP of 10 articles. Find profit%.

13. A dealer sells 900g as 1kg at the same price. Find effective gain%.

14. Marked price is 40% above CP and discount is 15%. Find profit%.

■ HARD

Solve without looking at the answer key. Mark one ✓ when done.

15. After a 20% discount, an article gives 25% profit. MP is what percent of CP?

16. Two articles are sold for 1200 each. One gives 20% gain and the other 20% loss. Find overall loss%.

17. An article costing 1200 is sold at 8% profit. Find SP.

18. Discount is 10% and gain is 26%. Find MP as % of CP.

19. An item sold at 10% loss and at 5% gain has a difference in SP of 180. Find CP.

20. Successive discounts are 20%, 15%, 10%. Find equivalent discount.

Profit & Loss — ANSWER KEY & MINI REVIEW

1. 20%	2. 720	3. 850	4. 16 2/3%	5. 850
6. 288	7. 1100	8. 800	9. 0%	10. 19%
11. 25%	12. 20%	13. 11 1/9%	14. 19%	15. 156.25%
16. 4%	17. 1296	18. 140%	19. 1200	20. 38.8%

■■ 20/20 TRACKER

Easy: ■1 ■2 ■3 ■4 ■5 ■6 ■7

Medium: ■8 ■9 ■10 ■11 ■12 ■13 ■14

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My mistake: _____

Correct idea/formula: _____

How I will avoid it next time: _____

Simple & Compound Interest

SI, CI, depreciation, compounding frequency and comparison.

■ SI

- $SI = P \times R \times T / 100$.
- Amount $A = P + SI = P(1 + RT/100)$.
- Rearrange the same formula to find P, R or T.

■ CI

- $A = P(1 + r/100)^n$; $CI = A - P$.
- Half-yearly: rate becomes $R/2$ and number of periods becomes $2T$.
- Quarterly: rate becomes $R/4$ and periods become $4T$.

■ DEPRECIATION & APPRECIATION

- Decrease $r\%$ repeatedly: $\text{final} = P(1 - r/100)^n$.
- Increase $r\%$ repeatedly: $\text{final} = P(1 + r/100)^n$.
- Do not use simple addition when compounding is repeated.

■ FAST COMPARISON

- For 2 years, $CI - SI = P(R/100)^2$.
- For 3 years, $CI - SI = P[3x^2 + x^3]$ where $x = R/100$.
- Use decimal multipliers for fast mental calculation.

FORMULA CARD

SI $PRT/100$

CI $P[(1 + r/100)^n - 1]$

Depreciation $P(1 - r/100)^n$

■ QUICK CHECK

SI on 10000 at 10% for 2 years?

Answer: 2000

■ QUICK CHECK

CI-SI for 10000 at 10% for 2 years?

Answer: 100

■ **SPEED TRICK**

Convert every compounding statement into 'rate per period + number of periods' before applying the formula.

■ **EXAM TRAP**

Do not halve the annual time instead of changing the rate/period correctly for half-yearly compounding.

Simple & Compound Interest — 20 QUESTION PRACTICE SET

Difficulty is progressive: Easy 1–7 • Medium 8–14 • Hard 15–20

■ EASY

Solve without looking at the answer key. Mark one ✓ when done.

1. Find SI on 5000 at 8% p.a. for 2 years.

2. $P=4000$, $R=10\%$, $T=3$ years. Find amount under SI.

3. $SI=2100$, $P=7000$, $T=3$ years. Find R .

4. $P=5000$, $R=8\%$, $SI=800$. Find T .

5. $SI=600$ at 12% for 1 year. Find P .

6. $P=10000$, $SI=1800$ in 3 years. Find R .

7. Find SI on 5000 at 12% for 1 year.

■ MEDIUM

Solve without looking at the answer key. Mark one ✓ when done.

8. Find amount on 12000 at 10% SI for 2 years.

9. $P=8000$ at 20% p.a. compounded half-yearly for 1.5 years. Find CI.

10. SI doubles a principal in 5 years. Find annual rate.

11. $P=20000$ at 10% CI for 2 years. Find CI.

12. Find CI–SI for 25000 at 8% p.a. for 2 years.

13. Amount is 13310 at 10% CI for 3 years. Find P.

14. $P=10000$ becomes 12100 in 2 years CI. Find R.

■ **HARD**

Solve without looking at the answer key. Mark one ✓ when done.

15. Find CI–SI on 50000 at 12% for 3 years.

16. Amount is 14520 at 10% CI for 2 years. Find P.

17. 8000 becomes 10648 in 3 years CI. Find annual rate.

18. A machine depreciates 10% yearly and is worth 72900 after 3 years. Find original value.

19. SI on a principal at 1% p.a. for 1 year is 500. Find P.

20. Half a principal earns 8% and half earns 12% for the same time. Find overall rate.

Simple & Compound Interest — ANSWER KEY & MINI REVIEW

1. 800	2. 5200	3. 10%	4. 2 years	5. 5000
6. 6%	7. 600	8. 14400	9. 2648	10. 20%
11. 4200	12. 160	13. 10000	14. 10%	15. 2246.40
16. 12000	17. 10%	18. 100000	19. 50000	20. 10%

■■ 20/20 TRACKER

Easy: ■1 ■2 ■3 ■4 ■5 ■6 ■7

Medium: ■8 ■9 ■10 ■11 ■12 ■13 ■14

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■ ERROR LOG — WRITE YOUR OWN PATTERN

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Ratio & Proportion

Ratios, mixtures, variation, proportional division and shortcuts.

■ RATIO BASICS

- $a:b$ means a/b . Multiplying or dividing both terms by the same non-zero number keeps the ratio unchanged.
- If total T is divided in $m:n$, shares are $mT/(m+n)$ and $nT/(m+n)$.
- Compound ratio: multiply corresponding terms.

■ PROPORTION

- $a:b=c:d$ means $ad=bc$.
- Third proportional to a,b is b^2/a . Fourth proportional to a,b,c is bc/a .
- Mean proportional between a,b is $\sqrt{ab}$.

■ VARIATION

- Direct: $x \propto y \Rightarrow x/y$ constant.
- Inverse: $x \propto 1/y \Rightarrow xy$ constant.
- Compound: $x \propto yz$ or $x \propto y/z$; write the constant equation before substituting.

■ EXAM SHORTCUTS

- Componendo: if $a/b=c/d$, then $(a+b)/b=(c+d)/d$.
- Dividendo: $(a-b)/b=(c-d)/d$.
- For linked ratios, equalize the common term using LCM before combining.

FORMULA CARD

Division Share = $mT/(m+n)$

Fourth proportional bc/a

Mean proportional $\sqrt{ab}$

■ QUICK CHECK

Divide 350 in 2:5.

Answer: 100, 250

■ QUICK CHECK

Mean proportional between 9 and 25?

Answer: 15

■ **SPEED TRICK**

For linked ratios, equalize the common term first. Do not add ratios that have different middle terms.

■ **EXAM TRAP**

A ratio has no units, but the quantities being compared must be in the same units.

Ratio & Proportion — 20 QUESTION PRACTICE SET

Difficulty is progressive: Easy 1–7 • Medium 8–14 • Hard 15–20

■ EASY

Solve without looking at the answer key. Mark one ✓ when done.

1. Simplify 18:27.

2. Divide 540 in the ratio 2:3. Find the two parts.

3. If $4:x=2:10$, find x .

4. Find fourth proportional to 3,5,9.

5. Find mean proportional between 4 and 25.

6. Divide 160 in ratio 3:2.

7. $A:B=1:4$ and $A+B=50$. Find A .

■ MEDIUM

Solve without looking at the answer key. Mark one ✓ when done.

8. $A:B=3:5$ and $B:C=10:7$. Find $A:B:C$.

9. Divide 1260 in ratio 2:3:4.

10. $x:y=5:6$ and $y:z=9:10$. Find $x:z$.

11. $a:b=2:3$. Find $(2a+2b):(3a+b)$.

12. 8 workers finish a job in 15 days. How long will 10 workers take?

13. Income:expenditure=5:4. Find savings as % of income.

14. Two numbers are in ratio 7:9 and differ by 26. Find them.

■ HARD

Solve without looking at the answer key. Mark one ✓ when done.

15. $A:B=4:5$, $B:C=6:7$, $C:D=8:9$. Find $A:D$.

16. Numbers are in ratio 2:3:5. Adding 20 to each gives 4:5:7. Find original numbers.

17. $a:b=3:4$, $b:c=5:6$, $c:d=7:8$. Find $a:d$.

18. $P=2Q$ and $Q=3R$. If $P+Q+R=660$, find P,Q,R .

19. x is directly proportional to y/z . If $x=12, y=8, z=4$, find x when $y=15, z=10$.

20. Milk:water=7:3 in 20 L. How much water must be added to make ratio 7:5?

Ratio & Proportion — ANSWER KEY & MINI REVIEW

1. 2:3	2. 216, 324	3. 20	4. 15	5. 10
6. 96, 64	7. 10	8. 6:10:7	9. 280, 420, 560	10. 3:4
11. 12:11	12. 12 days	13. 20%	14. 91, 117	15. 64:105
16. 20, 30, 50	17. 35:64	18. 396, 198, 66	19. 9	20. 4 L

■■ 20/20 TRACKER

Easy: ■1 ■2 ■3 ■4 ■5 ■6 ■7

Medium: ■8 ■9 ■10 ■11 ■12 ■13 ■14

Hard: ■15 ■16 ■17 ■18 ■19 ■20

■ ERROR LOG — WRITE YOUR OWN PATTERN

Question no.: _____ Topic: _____

My mistake: _____

Correct idea/formula: _____

How I will avoid it next time: _____

Average

Core average, weighted/combined average, replacement and speed.

■ CORE CONCEPT

- Average = total/count.
- Total = average×count.
- When one value changes by d, average changes by d/n for n observations.

■ COMBINED & WEIGHTED

- Combined average = $(n_1a_1+n_2a_2)/(n_1+n_2)$.
- Weighted average is used when observations have different frequencies/weights.
- Average of equally spaced consecutive numbers = middle value.

■ REPLACEMENT

- If one item x is replaced by y among n values, new average = old average + $(y-x)/n$.
- For multiple replacements, add all changes before dividing by n.

■ AVERAGE SPEED

- Average speed = total distance/total time.
- For equal distances at speeds a,b: average speed = $2ab/(a+b)$.
- Never take the ordinary average of speeds unless the time intervals are equal.

FORMULA CARD

Average Total/n

Combined $(n_1a_1+n_2a_2)/(n_1+n_2)$

Equal-distance speed $2ab/(a+b)$

■ QUICK CHECK

Average of 5 numbers is 18. Total?

Answer: 90

■ QUICK CHECK

Equal distances at 30 and 60 km/h. Average speed?

Answer: 40 km/h

■ SPEED TRICK

Think 'total' first. Most average questions become one line after converting average to total.

■ EXAM TRAP

Do not average averages unless group sizes are equal or weights are included.

Average — 20 QUESTION PRACTICE SET

Difficulty is progressive: Easy 1–7 • Medium 8–14 • Hard 15–20

■ EASY

Solve without looking at the answer key. Mark one ✓ when done.

1. Find average of 10,20,30,40,50.

2. Average of 8 numbers is 15. Find total.

3. Average of 5 numbers is 90. Find total.

4. Average of 4 numbers is 6.5. Find total.

5. Average of 7 numbers is 34. Find total.

6. Average of 10 numbers is 20. If one number 15 is removed, find new average.

7. Average of 12 numbers is 12. Find total.

■ MEDIUM

Solve without looking at the answer key. Mark one ✓ when done.

8. Average of 15 observations is 40. Find total.

9. Average of 20 is 18 and average of 5 is 42. Find combined average.

10. Average of 10 numbers is 12. If each is increased by 3, new average?

11. Average of 12 is 20. A value recorded as 23 should be 32. Find corrected average.

12. Average income of 10 people is 20000. One new person with income 16000 joins. Find new average.

13. Find average of first 20 positive even numbers.

14. A travels equal distances at 40 and 60 km/h. Find average speed.

15. Average of 10 scores is 50. Replace 72 by 42. Find new average.

■ **HARD**

Solve without looking at the answer key. Mark one ✓ when done.

16. $A+B+C=216$ and $B+C+D=225$. If $A=66$, find D.

17. Average of n students is 16. Adding one student with score 40 raises average to 17. Find n .

18. Average of 11 numbers is 48. Average of first 6 is 42 and last 6 is 54. Find the 6th number.

19. Average of 20 scores is 45. What score makes average 47 after adding one more score?

20. Replacing one weight by another raises average by 1.5 in a group of 30. Find increase in total weight.

Average — ANSWER KEY & MINI REVIEW

1. 30	2. 120	3. 450	4. 26	5. 238
6. 20 5/9	7. 144	8. 600	9. 22.8	10. 15
11. 20.75	12. 19636.36	13. 21	14. 48 km/h	15. 47
16. 75	17. 23	18. 48	19. 87	20. 45

■■ 20/20 TRACKER

Easy: ■1 ■2 ■3 ■4 ■5 ■6 ■7

Medium: ■8 ■9 ■10 ■11 ■12 ■13 ■14

Hard: ■15 ■16 ■17 ■18 ■19 ■20

■ ERROR LOG — WRITE YOUR OWN PATTERN

Question no.: _____ Topic: _____

My mistake: _____

Correct idea/formula: _____

How I will avoid it next time: _____

Time & Work

Rates, efficiency, LCM method, pipes and changing teams.

■ RATE MODEL

- If A finishes in x days, one-day work = $1/x$.
- Together, add rates: $1/T = 1/a + 1/b + \dots$
- Efficiency is inverse of time: higher efficiency means fewer days.

■ LCM METHOD

- Take total work as LCM of individual days. Daily work = total work/days.
- This avoids fractions and is excellent for exam speed.
- Worker-days-hours: Work $\propto$ workers $\times$ days $\times$ hours when efficiency is unchanged.

■ PIPES & CISTERNS

- Filling pipe = positive rate; emptying pipe = negative rate.
- Net rate = sum of all rates. If net rate is r , time = $1/r$ for one tank.
- Convert all rates to the same unit before adding.

■ ALTERNATING WORK

- For A, B, C working in cycles, calculate work done in one complete cycle first.
- Then find the remaining fraction and identify who works next.
- For wages, divide payment in proportion to actual work done.

FORMULA CARD

Rate $1/\text{time}$

Together Add rates

Worker-days $W \propto \text{workers} \times \text{days} \times \text{hours}$

■ QUICK CHECK

A=10 days, B=15 days. Together?

Answer: 6 days

■ QUICK CHECK

A does $\frac{1}{4}$ work in 5 days. Full time?

Answer: 20 days

■ SPEED TRICK

Think rate, not days. If someone is twice as efficient, their time is half.

■ EXAM TRAP

Adding times is wrong for people working together. Add rates.

Time & Work — 20 QUESTION PRACTICE SET

Difficulty is progressive: Easy 1–7 • Medium 8–14 • Hard 15–20

■ EASY

Solve without looking at the answer key. Mark one ✓ when done.

1. A finishes a job in 10 days. Find one-day work.

2. A finishes in 12 days, B in 18 days. Find together time.

3. A does $\frac{1}{4}$ work in 15 days. Find full time.

4. A and B together do a job in 4 days. Find combined daily rate.

5. 12 workers finish in 20 days. How many worker-days?

6. A and B together finish in 10 days. A alone takes 15 days. Find B alone.

7. A does $\frac{2}{5}$ of a job in 24 days. Find full time.

8. A does a job in 18 days. How much work in 6 days?

■ MEDIUM

Solve without looking at the answer key. Mark one ✓ when done.

9. A+B finish in 12 days. A alone in 20 days. Find B alone.

10. A,B,C take 12,18,36 days. Find together time.

11. 8 men do the same work as 12 women in the same time. Find man:woman efficiency ratio.

12. A does $\frac{1}{3}$ work in 5 days; B does $\frac{1}{4}$ in 6 days. Find efficiency ratio A:B.

13. 10 workers working 8 h/day for 15 days produce a job. Find worker-hours.

14. A+B=8 days, B+C=12 days, C+A=10 days. Find time if all work together.

15. A is 50% more efficient than B. Together they finish in 12 days. Find A's time.

■ **HARD**

Solve without looking at the answer key. Mark one ✓ when done.

16. A takes 20 days, B 30 days. They work 5 days; C finishes the remaining work in 6 days. Find C's time alone.

17. 15 men finish a job in 24 days. After 8 days, efficiency per man rises 25%. How many men can leave and still finish on time?

18. A,B,C work one day each repeatedly; their times alone are 10,15,30 days. Find total time.

19. A fills in 12 h, B in 18 h, C empties in 36 h. Find time together.

20. A+B finish in 6 days. A alone in 10 days. B works alone for 4 days, then A joins. Find total time.

Time & Work — ANSWER KEY & MINI REVIEW

1. $\frac{1}{10}$	2. 7.2 days	3. 60 days	4. $\frac{1}{4}$	5. 240
6. 30 days	7. 60 days	8. $\frac{1}{3}$	9. 30 days	10. 6 days
11. 3:2	12. 8:5	13. 1200	14. 240/37 days	15. 20 days
16. $\frac{72}{7}$ days	17. 3 men	18. 15 days	19. 9 h	20. $11\frac{1}{3}$ days

■■ 20/20 TRACKER

Easy: ■1 ■2 ■3 ■4 ■5 ■6 ■7

Medium: ■8 ■9 ■10 ■11 ■12 ■13 ■14

Hard: ■15 ■16 ■17 ■18 ■19 ■20

■ ERROR LOG — WRITE YOUR OWN PATTERN

Question no.: _____ Topic: _____

My mistake: _____

Correct idea/formula: _____

How I will avoid it next time: _____

Time, Speed & Distance

Relative speed, trains, boats, average speed and catch-up.

■ CORE FORMULA

- Speed=Distance/Time; Distance=Speed×Time; Time=Distance/Speed.
- 1 m/s=18/5 km/h; 1 km/h=5/18 m/s.
- Use consistent units before calculating.

■ RELATIVE SPEED

- Same direction: relative speed= $|u-v|$.
- Opposite directions: relative speed= $u+v$.
- Catch-up time = head start distance / relative speed.

■ TRAINS

- Train crossing a point: time=train length/speed.
- Train crossing a platform: time=(train length+platform length)/speed.
- Two trains crossing: use sum of lengths and relative speed.

■ BOATS & STREAMS

- Downstream speed= $u+v$; upstream speed= $u-v$.
- Still-water speed $u=(\text{down}+\text{up})/2$; stream speed $v=(\text{down}-\text{up})/2$.
- Equal-distance average speed again uses $2ab/(a+b)$.

FORMULA CARD

Core $D=ST$

Conversion 1 m/s=18/5 km/h

Relative same: difference; opposite: sum

■ QUICK CHECK

72 km/h in m/s?

Answer: 20 m/s

■ QUICK CHECK

Train length 200m at 20m/s crosses pole in?

Answer: 10 s

■ SPEED TRICK

Make a unit check your first step. Most avoidable TSD mistakes are km/h vs m/s.

■ EXAM TRAP

For equal distances, average speed is not the arithmetic mean.

Time, Speed & Distance — 20 QUESTION PRACTICE SET

Difficulty is progressive: Easy 1–7 • Medium 8–14 • Hard 15–20

■ EASY

Solve without looking at the answer key. Mark one ✓ when done.

1. Convert 54 km/h to m/s.

2. A car travels at 60 km/h for 2 h. Find distance.

3. A 90 km trip at 30 km/h takes how long?

4. A train travels 200 m at 20 m/s. Find time.

5. Convert 25 m/s to km/h.

6. A 72 km trip at 36 km/h takes how long?

7. A runner covers 150 m in 10 s. Find speed.

■ MEDIUM

Solve without looking at the answer key. Mark one ✓ when done.

8. A 300 m train at 36 km/h crosses a pole. Find time.

9. Equal distances are covered at 40 and 60 km/h. Find average speed.

10. Speed rises by 25%. Find percentage decrease in time for same distance.

11. Downstream speed=15, upstream=10 km/h. Find still-water and stream speeds.

12. A train at 54 km/h crosses a 300 m bridge in 30 s. Find train length.

13. Two cyclists move opposite at 6 and 4 m/s. How fast do they separate?

14. A 60 km trip is completed in 1.6 h. Find average speed.

15. A train crosses a pole in 12 s and a 180 m platform in 24 s. Find speed and train length.

■ **HARD**

Solve without looking at the answer key. Mark one ✓ when done.

16. Two trains of total length 400 m move opposite at 90 and 60 km/h. Find crossing time.

17. A person increases speed from 12 to 15 km/h and saves 30 min. Find distance.

18. Upstream=6 km/h, downstream=12 km/h. Find still-water and stream speeds.

19. B starts 20 min before A at 50 km/h. A travels at 60 km/h. Find A's catch-up time after A starts.

20. A 72 km/h train crosses a pole in 8 s and a platform in 28 s. Find platform length.

Time, Speed & Distance — ANSWER KEY & MINI REVIEW

1. 15 m/s	2. 120 km	3. 3 h	4. 10 s	5. 90 km/h
6. 2 h	7. 15 m/s	8. 30 s	9. 48 km/h	10. 20%
11. 12.5 km/h, 2.5 km/h	12. 150 m	13. 10 m/s	14. 37.5 km/h	15. 15 m/s; 180 m
16. 48 s	17. 30 km	18. 9 km/h, 3 km/h	19. 100 min	20. 400 m

■■ 20/20 TRACKER

Easy: ■1 ■2 ■3 ■4 ■5 ■6 ■7

Medium: ■8 ■9 ■10 ■11 ■12 ■13 ■14

Hard: ■15 ■16 ■17 ■18 ■19 ■20

■ ERROR LOG — WRITE YOUR OWN PATTERN

Question no.: _____ Topic: _____

My mistake: _____

Correct idea/formula: _____

How I will avoid it next time: _____

Probability

Sample space, complements, cards, dice, conditional and dependent events.

■ CORE CONCEPT

- Probability = number of favourable outcomes / total outcomes when outcomes are equally likely.
- $0 \leq P(E) \leq 1$.
- Complement: $P(\text{not } E) = 1 - P(E)$.

■ ADDITION & MULTIPLICATION

- $P(A \text{ or } B) = P(A) + P(B) - P(A \text{ and } B)$.
- Mutually exclusive events have $P(A \text{ and } B) = 0$.
- Independent events: $P(A \text{ and } B) = P(A)P(B)$.

■ DEPENDENT EVENTS

- Without replacement, the denominator and sometimes numerator change after each draw.
- Conditional probability $P(A|B) = P(A \cap B) / P(B)$.
- 'At least one' is often fastest by complement: $1 - P(\text{none})$.

■ STANDARD SETS

- Cards: 52 cards, 4 suits, 13 ranks; face cards J, Q, K = 12.
- Two dice: 36 equally likely ordered outcomes.
- n coins have 2^n outcomes.

FORMULA CARD

Basic favourable/total

Complement $1 - P(E)$

Union $P(A) + P(B) - P(A \cap B)$

■ QUICK CHECK

One die: $P(\text{prime})$?

Answer: $1/2$

■ QUICK CHECK

At least one head in two tosses?

Answer: $\frac{3}{4}$

■ SPEED TRICK

Look for the complement when the wording says 'at least one', 'none', or 'not'.

■ EXAM TRAP

'Or' usually means union; 'and' usually means intersection. Check overlap before adding.

Probability — 20 QUESTION PRACTICE SET

Difficulty is progressive: Easy 1–7 • Medium 8–14 • Hard 15–20

■ EASY

Solve without looking at the answer key. Mark one ✓ when done.

1. Probability of getting a head on a fair coin.

2. Probability of rolling a 6 on a fair die.

3. Probability of an even number on a die.

4. Probability of drawing an ace from a standard deck.

5. Probability of not getting a 6 on a die.

6. Probability of two heads in two fair coin tosses.

7. Probability of choosing a vowel from A,E,I,O,U.

8. Probability of drawing a king from a deck.

■ MEDIUM

Solve without looking at the answer key. Mark one ✓ when done.

9. Two dice: probability that sum is at least 10.

10. Three coins: probability of exactly two heads.

11. Probability of drawing a face card from a deck.

12. A bag has 3 red and 7 blue balls. Probability of red.

13. A bag has 4 red and 6 blue. Two reds without replacement?

14. On a die, probability of prime or even.

15. Two dice: probability that product is even.

■ HARD

Solve without looking at the answer key. Mark one ✓ when done.

16. Draw 3 cards. Probability all are aces.

17. Two balls drawn from 10 (4 red, 6 blue). Probability at least one red.

18. Two dice: probability that the sum is prime.

19. Choose 2 from 5 men and 4 women. Probability both are women.

20. A box has 3 defective and 7 good items. Two selected. Probability exactly one defective.

Probability — ANSWER KEY & MINI REVIEW

1. $\frac{1}{2}$	2. $\frac{1}{6}$	3. $\frac{1}{2}$	4. $\frac{1}{13}$	5. $\frac{5}{6}$
6. $\frac{1}{4}$	7. 1	8. $\frac{1}{13}$	9. $\frac{1}{6}$	10. $\frac{3}{8}$
11. $\frac{3}{13}$	12. $\frac{3}{10}$	13. $\frac{2}{15}$	14. $\frac{5}{6}$	15. $\frac{3}{4}$
16. $\frac{1}{5525}$	17. $\frac{13}{15}$	18. $\frac{5}{12}$	19. $\frac{1}{6}$	20. $\frac{7}{15}$

■■ 20/20 TRACKER

Easy: ■1 ■2 ■3 ■4 ■5 ■6 ■7

Medium: ■8 ■9 ■10 ■11 ■12 ■13 ■14

Hard: ■15 ■16 ■17 ■18 ■19 ■20

■ ERROR LOG — WRITE YOUR OWN PATTERN

Question no.: _____ Topic: _____

My mistake: _____

Correct idea/formula: _____

How I will avoid it next time: _____

Permutation & Combination

Counting, restrictions, circular arrangements, repetition and distributions.

■ WHEN TO USE

- Permutation: order matters. Combination: order does not matter.
- $nPr = n!/(n-r)!$; $nCr = n!/[r!(n-r)!]$.
- $nCr = nC(n-r)$.

■ REPETITION

- Arrangements of n objects with repeats a, b, c : $n!/(a!b!c!)$.
- Length- r arrangements with repetition allowed from n symbols = n^r .
- Selections with repetition from n types choosing r = $C(n+r-1, r)$.

■ RESTRICTIONS

- Together: make a block, arrange blocks, then multiply by internal arrangements.
- Circular arrangements of n distinct people: $(n-1)!$.
- For 'not together', use total-together when easier.

■ DISTRIBUTION

- Non-negative solutions $x_1 + \dots + x_k = n$: $C(n+k-1, k-1)$.
- Positive solutions: $C(n-1, k-1)$.
- For identical objects in distinct boxes, stars-and-bars is the standard tool.

FORMULA CARD

nPr $n!/(n-r)!$

nCr $n!/[r!(n-r)!]$

Circle $(n-1)!$

■ QUICK CHECK

6P2?

Answer: 30

■ QUICK CHECK

8C2?

Answer: 28

■ SPEED TRICK

Ask one question: 'Does order matter?' If yes, permutation; if no, combination.

■ EXAM TRAP

'Together' does not mean just multiply nPr ; form a block and count internal orders.

Permutation & Combination — 20

QUESTION PRACTICE SET

Difficulty is progressive: Easy 1–7 • Medium 8–14 • Hard 15–20

■ EASY

Solve without looking at the answer key. Mark one ✓ when done.

1. Find $5!$.

2. Find $6P2$.

3. Find $7C2$.

4. Arrange 4 distinct books on a shelf.

5. Choose 3 people from 8.

6. Arrange 3 distinct letters from 4.

7. How many ways to arrange 3 people in a row?

■ MEDIUM

Solve without looking at the answer key. Mark one ✓ when done.

8. How many distinct arrangements of LEVEL?

9. Arrange 5 distinct people in a row.

10. Arrange 5 distinct books with 2 specific books together.

11. Arrange 5 people around a circle.

12. Choose 2 men from 5 and 3 women from 6.

13. How many 4-digit numbers can be made from 0–9 without repetition?

14. Arrange 4 objects if 2 specified must be together.

■ **HARD**

Solve without looking at the answer key. Mark one ✓ when done.

15. How many distinct arrangements of MISSISSIPPI?

16. Arrange 8 people around a circle if two specified people sit together.

17. Choose exactly 2 aces from a 52-card deck.

18. How many 5-digit numbers divisible by 5 have distinct digits?

19. 10 identical balls into 4 distinct boxes, boxes may be empty.

20. Positive integer solutions $x+y+z=10$.

Permutation & Combination — ANSWER KEY & MINI REVIEW

1. 120	2. 30	3. 21	4. 24	5. 56
6. 24	7. 6	8. 30	9. 120	10. 48
11. 24	12. 200	13. 4536	14. 48	15. 34650
16. 1440	17. 6768	18. 5712	19. 286	20. 36

■■ 20/20 TRACKER

Easy: ■1 ■2 ■3 ■4 ■5 ■6 ■7

Medium: ■8 ■9 ■10 ■11 ■12 ■13 ■14

Hard: ■15 ■16 ■17 ■18 ■19 ■20

■ ERROR LOG — WRITE YOUR OWN PATTERN

Question no.: _____ Topic: _____

My mistake: _____

Correct idea/formula: _____

How I will avoid it next time: _____

Data Interpretation

Tables, charts, percentages, ratios, averages, profit and mixed DI.

■ READ FIRST

- Check units, total, scale and headings before calculating.
- Identify exactly what is being asked: difference, ratio, percentage, average, share or growth.
- Estimate before exact calculation to catch impossible answers.

■ TABLES & GRAPHS

- For a table, compute only the rows/columns needed.
- Line charts: compare trends, changes and percentage growth.
- Bar charts: compare magnitudes and differences.

■ PIE CHART

- Sector angle = $\text{value}/\text{total} \times 360^\circ$.
- Value = $\text{angle}/360^\circ \times \text{total}$.
- Percentage share = $\text{angle}/3.6$.

■ DI SHORTCUTS

- Ratio questions can often be solved directly from chart values without finding the total.
- For percentage change, use $(\text{new}-\text{old})/\text{old} \times 100$.
- For multi-year growth, distinguish total percentage growth from annual compound growth.

FORMULA CARD

Percentage $\text{change}/\text{original} \times 100$

Pie value $\text{angle}/360 \times \text{total}$

Pie % $\text{angle}/3.6$

■ QUICK CHECK

Pie sector 72° = what percentage?

Answer: 20%

■ QUICK CHECK

Values 40,60,80 average?

Answer: 60

■ SPEED TRICK

Estimate first, calculate second. It prevents unit and decimal mistakes.

■ EXAM TRAP

Never read a graph without checking scale, units and whether values are absolute or percentages.

Data Interpretation — 20 QUESTION PRACTICE SET

Difficulty is progressive: Easy 1–7 • Medium 8–14 • Hard 15–20

■ EASY

Solve without looking at the answer key. Mark one ✓ when done.

1. Table: Mon=20, Tue=30, Wed=25, Thu=35, Fri=40. Find total.

2. Using the same table, which day has maximum value?

3. Using the same table, find average.

4. If each unit earns 5, find Friday revenue.

5. Monday's share of total as a percentage?

6. Difference between Thursday and Tuesday values?

7. Ratio of Wednesday to Monday values.

■ MEDIUM

Solve without looking at the answer key. Mark one ✓ when done.

8. A value rises from 120 to 210. Find percentage increase.

9. A company sells 100, 120, 140, 160 units over four years. Find average.

10. Revenue=1500, cost=1070. Find profit.

11. Profit rises from 100 to 120. Find percentage increase.

12. 30 of 100 students choose option A. Find percentage.

13. Maximum and minimum chart values are 80 and 40. Find ratio.

14. Five departments have values 40,50,60,60,90. Find total.

15. Sales rise 240→450 over 3 years. Find approximate CAGR.

■ HARD

Solve without looking at the answer key. Mark one ✓ when done.

16. A:B=3:4 and B:C=5:6. Total A+B+C=590. Find C.

17. A chart value rises from 50 to 100. Find percentage increase.

18. Revenue rises 500→600 and cost 350→385. Find percentage increase in profit.

19. Largest category is 40% of 2400 and smallest is 15%. Find difference.

20. A survey has 1000 students; 40% are girls. Find number of boys.

Data Interpretation — ANSWER KEY & MINI REVIEW

1. 150	2. Friday	3. 30	4. 200	5. 13.33%
6. 5	7. 5:4	8. 75%	9. 130	10. 430
11. 20%	12. 30%	13. 2:1	14. 300	15. 23.3%
16. 240	17. 100%	18. 43.33%	19. 600	20. 600

■■ 20/20 TRACKER

Easy: ■1 ■2 ■3 ■4 ■5 ■6 ■7

Medium: ■8 ■9 ■10 ■11 ■12 ■13 ■14

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■ ERROR LOG — WRITE YOUR OWN PATTERN

Question no.: _____ Topic: _____

My mistake: _____

Correct idea/formula: _____

How I will avoid it next time: _____

FINAL REVISION BOARD

Use this last section for rapid revision before an aptitude test.

NUMBER SYSTEM

Divisibility • HCF/LCM • remainders • cycles • factorial • phi

PERCENTAGE

Multiplier method • successive change • reverse % • comparison

PROFIT & LOSS

CP/SP/MP • profit/loss base • discount • markup • short-weight

SI & CI

$SI = \frac{P \times R \times T}{100}$ • $CI = P[(1+r)^n - 1]$ • compounding • depreciation

RATIO

Equalize common terms • division • variation • proportionality

AVERAGE

Total = $n \times \text{avg}$ • combined/weighted • replacement • equal-distance speed

WORK

Rate = $\frac{1}{\text{time}}$ • add rates • LCM method • worker-days • pipes

TSD

$D = ST$ • relative speed • trains • boats • catch-up

PROBABILITY

Favourable/total • complement • union • independent/dependent

P&C;

Order matters? • nPr • nCr • blocks • circles • stars & bars

DI

Read scale/units • percentage • ratio • average • estimate first

■ MASTER ERROR LOG

1. Topic: _____ Question: _____ Mistake: _____
2. Topic: _____ Question: _____ Mistake: _____
3. Topic: _____ Question: _____ Mistake: _____
4. Topic: _____ Question: _____ Mistake: _____
5. Topic: _____ Question: _____ Mistake: _____
6. Topic: _____ Question: _____ Mistake: _____